Experiment 9

AI and Data Analytics Using Zoho Zia in Zoho Creator

Aim

To implement AI and data analytics using Zoho Zia in Zoho Creator to analyze vehicle, driver, passenger, route and trip data, identify trends and generate insights for transport operations.

Procedure

1. Log in to Zoho Creator and choose the option to create an application with Zia (AI app builder).
2. Describe the requirement as a transport management application. Zia suggests the use cases and a data model (for example Vehicle Details, Driver Details, Passenger Details, Route Details and Trip Scheduling).
3. Review the suggested use cases and data model, then click Create Application.
4. Open the generated application and review the Dashboard with the Zia-generated analytics: total vehicles, drivers, passengers and trips, and charts of vehicle status, vehicle type, driver status and trip status.
5. Add sample vehicles, drivers, passengers, routes and trips so that the dashboard has data to analyze.
6. Open Edit this application and use the Zia Build Agent to modify or extend the application through instructions (for example, add a Fuel Log form).
7. Open the Vehicle Details Report and review the status of each vehicle (Available, On Trip, Under Maintenance) with insurance expiry dates.
8. Study the charts and reports to identify transport demand trends.

Sample Prompts Given to Zia

- "Create an application to manage vehicles, drivers, passengers, routes and trip scheduling with vehicle status and insurance expiry dates."
- "Add a dashboard card showing the total number of vehicles, drivers, passengers and trips."
- "Add a report that lists vehicles whose insurance expires within 30 days."

Process Diagram

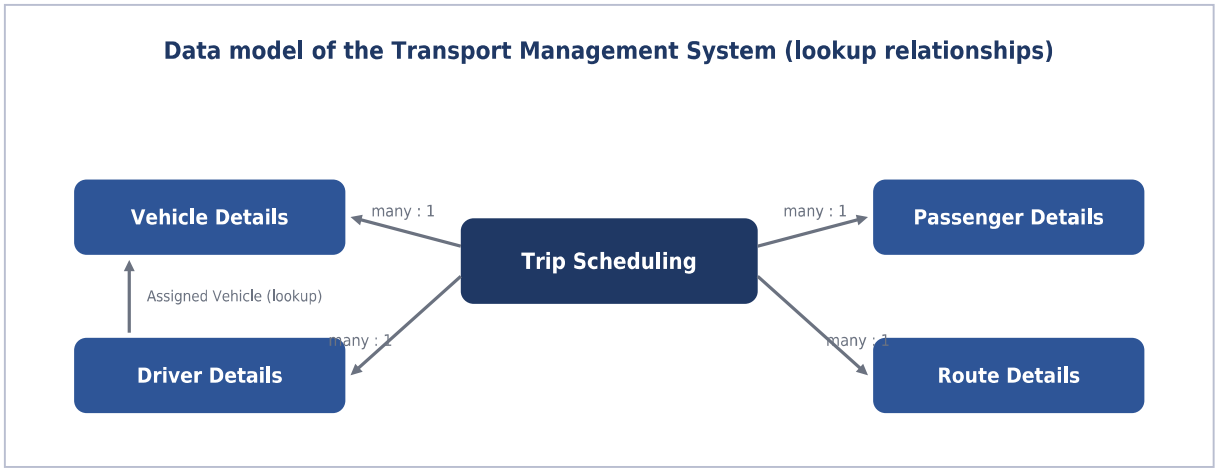


Figure 9.1: Diagram of the data model suggested by Zia (illustrative diagram, not a screenshot).

Dashboard Metrics Generated for the Sample Data

Sample data for this lab record. Values shown are the expected output for these sample records.

Metric	Value
KPI cards	Total Vehicles 5, Total Drivers 5, Total Passengers 5, Total Trips 5
Vehicle status (pie chart)	Available 3, On Trip 1, Under Maintenance 1
Vehicle type (donut chart)	Bus 2, Mini Bus 1, Van 1, Car 1
Driver status (bar chart)	Active 4, On Leave 1
Trip status (column chart)	Scheduled 1, In Progress 1, Completed 2, Cancelled 1

Insurance Check Records (Sample)

Vehicle	Insurance Status	Remark
TN 04 GH 3456	Expires within 30 days	Alert email sent to the administrator (expected)
TN 01 AB 1234	Valid	No action required
TN 02 CD 5678	Valid	No action required

Trends Identified

- Chennai – Vellore and Chennai – Kanchipuram are the most used routes in the sample data.
- Buses form the largest vehicle type, and one bus under maintenance reduces available seating capacity.
- Early insurance expiry alerts support timely renewal and avoid unplanned vehicle downtime.

Result

AI and analytics using Zia helped generate a transport management application, visualize important metrics such as vehicle status and trip status, and identify transport demand trends.

Experiment 10

Performance Optimization of the Transport Management System

Aim

To optimize the performance of the Transport Management System in Zoho Creator by improving form loading speed, reducing unnecessary database operations, optimizing Deluge scripts and ensuring efficient report generation.

Procedure

1. Log in to Zoho Creator and open the Transport Management System application.
2. Review the Vehicle_Details, Driver_Details and Trip_Scheduling forms and remove unnecessary fields, scripts and workflows.
3. Optimize Deluge scripts by avoiding repeated database queries and unnecessary loops.
4. Use criteria-based record fetching to retrieve only the required vehicles and trips.
5. Optimize reports by displaying only essential columns and applying suitable filters.
6. Reduce unnecessary workflow executions and avoid duplicate email notifications.
7. Test the application after the changes to compare loading speed and responsiveness.
8. Verify that all transport features work correctly after optimization.

Optimization Actions

Area	Before	After
Form	Duplicate and unused fields (extra phone, unused media fields) on the form	Only required fields kept; fields grouped in sections
Deluge fetch	Fetch all vehicles, then test the insurance date inside a loop	Criteria-based fetch: only vehicles expiring within 30 days are fetched
Reports	All columns shown, no filters	Essential columns only; filters on Trip Status and Vehicle Status
Workflows	Status workflow runs on every edit of a trip	Runs only when Trip Status changes to In Progress or Completed

Deluge Optimization Example

BEFORE:

```
for each v in Vehicle_Details
{
    if(v.Insurance_Expiry_Date <= zoho.currentdate.addDay(30))
    {
        sendmail
        [
            from : zoho.adminuserid
            to : zoho.adminuserid
            subject : "Insurance Expiry Alert - " + v.Vehicle_Number
            message : "..."
        ]
    }
}
```